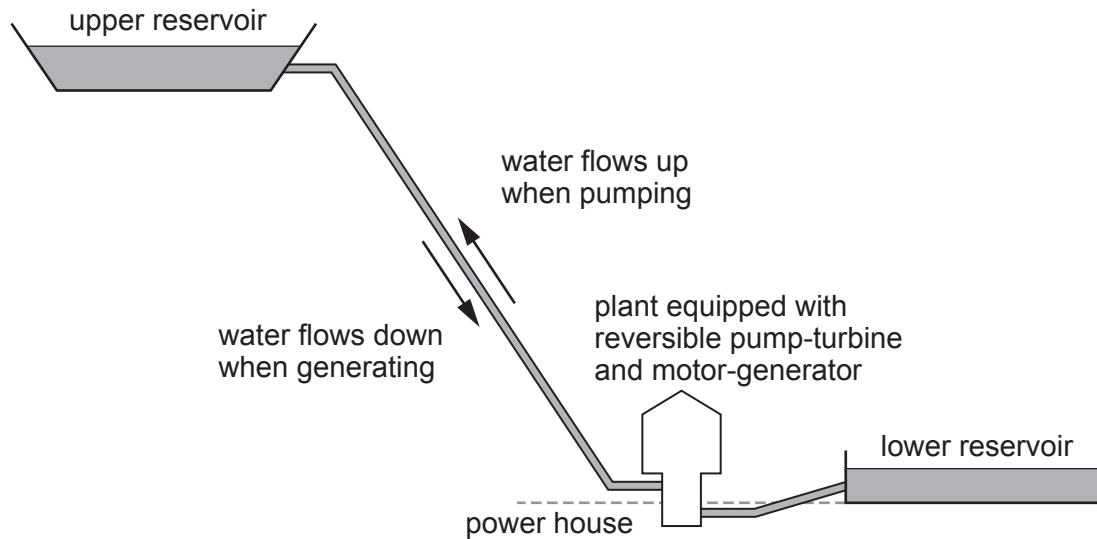


6. Dinorwig is a pumped storage hydroelectric power station in North Wales. Water is stored in an upper reservoir to be released when electricity is required. Water is then pumped back to the upper reservoir.



At full capacity Dinorwig produces 1800 MW of power for 6 hours. This is transferred to the National Grid via a step-up transformer.

- (a) (i) Explain why a step-up transformer is used.

[2]

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- (ii) Use an equation from page 2 to calculate the output current from the transformer at full capacity if the output voltage is 400 kV.

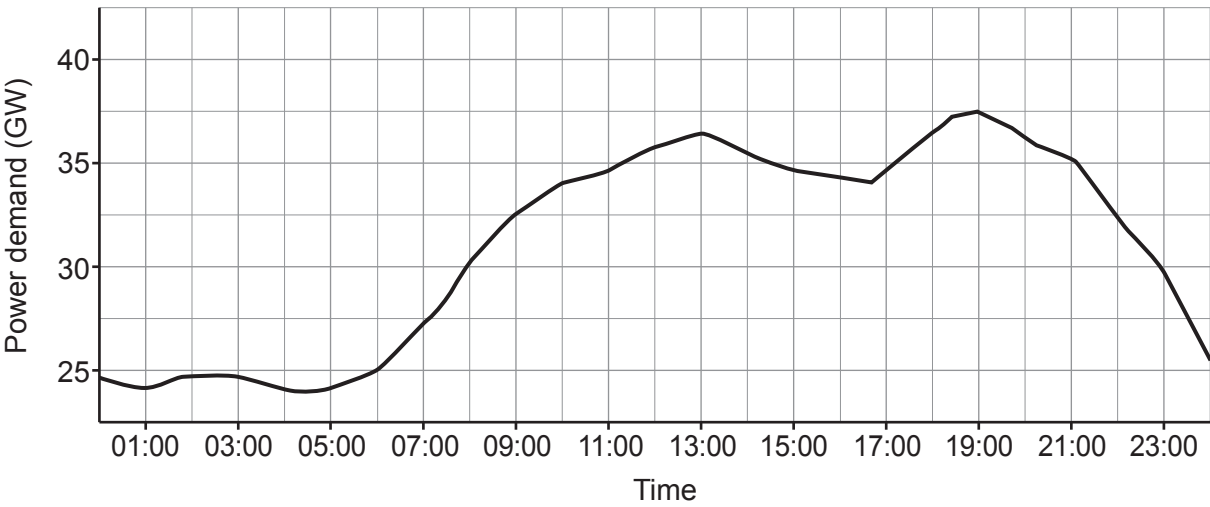
[3]

output current = ..... A



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only

(b) The graph shows how power demand for electricity varied throughout the day for one day in March.



Use the graph to state between which times water will be pumped back to the upper reservoir.  
Explain your answer.

[2]

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END OF PAPER

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| 7 |

